

CH 1101: Quantum Chemistry and Spectroscopy

Instructor-in-charge:

Dr. Shampa Raghunathan

Email ID: shampa.raghunathan@mahindrauniversity.edu.in

Co-instructor:

Dr. Venkatraman NV

Email ID: venkatraman.nv@mahindrauniversity.edu.in

Timings:

Batch 01 (AI, CSE, ICSE)
Monday @02:35 - 03:30 PM, [Class Room: ELT04, SF](#)
Tuesday @03:35 - 04:30 PM, [Class Room: ELT04, SF](#)

Batch 02 (ECE, ECM, VLSI, AERO, NT)
Monday @10:35 - 11:30 AM, [Class Room: ELT02, GF](#)
Tuesday @10:35 - 11:30 AM, [Class Room: ELT04, SF](#)

Tutors:

1. Dr. Sonu Kumar
2. Dr. Sourav Pradhan
3. Dr. Basudev Maity
4. Dr. Venkatraman NV
5. Dr. Shampa Raghunathan

Grading Pattern:

Minor I Examination	15%
Minor II Examination	15%
End-semester Examination	40%
Internal Assessment	10%
Laboratory	20%

Suggested Reading:

Text book:

1. Atkin's Physical Chemistry; 10th edition; Peter Atkins and Julio De Paula

Reference books:

2. Fundamentals of Molecular Spectroscopy; 4th edition; Colin N. Banwell and Elaine M. McCash

3. Applications of absorption spectroscopy of organic compounds; John R. Dyer

Book 1: Chapters 7, 8, 9 and 11; topics communicated during lecture

CH 1101: Quantum Chemistry and Spectroscopy (L-T-P: 2-1-2; Credits: 4) - Fall 2026 Schedule					
Lecture Timings:					
Week	Date	Day	Lec.	Lecture Topics	Tutorials
1	10-Aug	Mon	1	Introduction to CH 1101: Quantum mechanics	Tutorial - 1
	11-Aug	Tue	2	Origins of quantum mechanics	
2	17-Aug	Mon	3	Origins of quantum mechanics	Tutorial - 2
	18-Aug	Tue	4	Principles of quantum mechanics	
3	24-Aug	Mon	5	Principles of quantum mechanics	Tutorial - 3
	25-Aug	Tue	6	Free particle	
4	31-Aug	Mon	7	Particle in one-dimensional box: Wavefunctions and probabilities	Tutorial - 4
	01-Sep	Tue	8	Particle in one-dimensional box: Energies	
5	07-Sep	Mon	9	Particle in a ring: Wavefunctions and probabilities	Tutorial - 5
	08-Sep	Tue	10	Particle in a ring: Energies	
5	Minor Examinations I (September 09 - 12)				
6	14-Sep	Mon		Holiday	Tutorial - 6
	15-Sep	Tue	11	H-atom (only solutions) and shapes of atomic orbitals	
7	21-Sep	Mon	12	The chemical bond: Molecular orbital theory (MOT)	Tutorial - 7
	22-Sep	Tue	13	The chemical bond: MOT & Symmetry	
8	28-Sep	Mon	14	The chemical bond: Homonuclear diatomic molecules	Tutorial - 8
	29-Sep	Tue	15	The chemical bond: Homonuclear diatomic molecules	
9	05-Oct	Mon	16	Fundamentals of molecular spectroscopy	Tutorial - 9
	06-Oct	Tue	17	Beer-Lambert's law, peak widths	
10	12-Oct	Mon	18	UV-Visible spectroscopy: Electronic transitions	Tutorial - 10
	13-Oct	Tue	19	UV-Visible spectroscopy of organic molecules	
11	Fall Break (October 19 - 23)				
12	26-Oct	Mon	20	Infrared spectroscopy: Introduction	Tutorial - 11
	27-Oct	Tue	21	Infrared spectroscopy: Normal modes	
12	Minor Examinations II (October 28 - 31)				
13	02-Nov	Mon	22	Infrared spectroscopy: Peak position and intensity	Tutorial - 12
	03-Nov	Tue	23	Infrared spectroscopy: Interpretation	
14	09-Nov	Mon		Holiday	Tutorial - 13
	10-Nov	Tue	24	Fundamentals of NMR spectroscopy, Chemical shift	
15	16-Nov	Mon	25	Spin-spin interactions	Tutorial - 14
	17-Nov	Tue	26	Structure determination using NMR and IR spectroscopy	
16	23-Nov	Mon	27	Structure determination using NMR and IR spectroscopy	Tutorial - 15
	24-Nov	Tue		Holiday	
17	30-Nov	Mon	28	Structure determination using NMR and IR spectroscopy	Tutorial - 15
	01-Dec	Tue	29	Structure determination using NMR and IR spectroscopy	
17	Buffer Days (December 02 - 03)				
17-19	End-semester Examinations (December 04 - 18)				

Course Objective:

-
1. To introduce the fundamental theories and postulates of quantum mechanics in view of chemical principles at the atomic scale.
-
2. To familiarize with theoretical and practical aspects of modern spectroscopic techniques for chemical analysis.

Course Outcome:

-
1. Importance of quantum mechanics for understanding the structure and properties (energetic aspects) of atoms and molecules.
-
2. Determine exact chemical structures of simple organic compounds using a combination of spectroscopic data.

Important Announcement:



1. Grading

- Missing Minor exams will not be given re-test. If a student misses it due to genuine and verifiable reason(s), she/he will be given weightage for the missed exam(s) on a pro-rate basis of performance in the End-sem exam.
- Missing End-sem will be awarded to ZERO (F-grade).

2. Attendance

- QR code scanning and it is mandatory! No mercy if attendance falls short.

3. Office hours

- Wednesday (2:00 - 3:00 PM, Office Place: Cabin no. 13, Faculty office 1, 1st floor, ECSOE building).